

Tabular Planning: Value Iteration

**Kianté Brantley
CS 1840 Fall 2026**

Announcements

HW1 is out!

HW0 is due Sept 17 (**this Thursday**) , 11:59 pm ET

Check the course website for Office Hours:
<https://harvard-cs-1840.github.io/2026/>

Recap: Fixed-point solution

Find the fixed point solution of $x^\star = f(x^\star)$, $x \in \mathbb{R}$

Recap: Fixed-point solution

Find the fixed point solution of $x^\star = f(x^\star)$, $x \in \mathbb{R}$

Start with some x_0 , set $x_{t+1} \Leftarrow f(x_t)$

Recap: Fixed-point solution

Find the fixed point solution of $x^\star = f(x^\star)$, $x \in \mathbb{R}$

Start with some x_0 , set $x_{t+1} \Leftarrow f(x_t)$

Suppose f is contraction, i.e., $\forall x, x', |f(x') - f(x)| \leq \gamma |x' - x|, \gamma \in [0, 1)$,
then $x_t \rightarrow x^\star$

Finishing Up From Last Lecture



1. Exact Policy Evaluation



2. Approximate Policy Evaluation via an Iterative Algorithm

Convergence of Iterative PE

Theorem:

Recall $\gamma \in [0, 1)$. After t iterations, we have:

$$\forall s, \left| V^t(s) - V^\pi(s) \right| \leq \gamma^t \left\| V^0 - V^\pi \right\|_\infty$$

Convergence of Iterative PE

Theorem:

Recall $\gamma \in [0, 1)$. After t iterations, we have:

$$\forall s, \left| V^t(s) - V^\pi(s) \right| \leq \gamma^t \left\| V^0 - V^\pi \right\|_\infty$$

$$\forall s, \left| V^{t+1}(s) - V^\pi(s) \right|$$

Convergence of Iterative PE

Theorem:

Recall $\gamma \in [0, 1)$. After t iterations, we have:

$$\forall s, \left| V^t(s) - V^\pi(s) \right| \leq \gamma^t \left\| V^0 - V^\pi \right\|_\infty$$

$$\forall s, \left| V^{t+1}(s) - V^\pi(s) \right|$$

$$= \left| r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \left(r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right) \right|$$

Convergence of Iterative PE

Theorem:

Recall $\gamma \in [0, 1)$. After t iterations, we have:

$$\forall s, \left| V^t(s) - V^\pi(s) \right| \leq \gamma^t \left\| V^0 - V^\pi \right\|_\infty$$

$$\forall s, \left| V^{t+1}(s) - V^\pi(s) \right|$$

$$= \left| r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \left(r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right) \right|$$

$$= \gamma \left| \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right|$$

Convergence of Iterative PE

Theorem:

Recall $\gamma \in [0, 1)$. After t iterations, we have:

$$\forall s, \left| V^t(s) - V^\pi(s) \right| \leq \gamma^t \left\| V^0 - V^\pi \right\|_\infty$$

$$\forall s, \left| V^{t+1}(s) - V^\pi(s) \right|$$

$$= \left| r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \left(r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right) \right|$$

$$= \gamma \left| \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right|$$

$$\leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} \left| V^t(s') - V^\pi(s') \right|$$

Convergence of Iterative PE

Theorem:

Recall $\gamma \in [0, 1)$. After t iterations, we have:

$$\forall s, \left| V^t(s) - V^\pi(s) \right| \leq \gamma^t \left\| V^0 - V^\pi \right\|_\infty$$

$$\begin{aligned} & \forall s, \left| V^{t+1}(s) - V^\pi(s) \right| \\ &= \left| r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \left(r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right) \right| \\ &= \gamma \left| \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right| \\ &\leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} \left| V^t(s') - V^\pi(s') \right| \\ &\leq \gamma \left\| V^t - V^\pi \right\|_\infty \end{aligned}$$

Convergence of Iterative PE

Theorem:

Recall $\gamma \in [0, 1)$. After t iterations, we have:

$$\forall s, \left| V^t(s) - V^\pi(s) \right| \leq \gamma^t \left\| V^0 - V^\pi \right\|_\infty$$

$$\forall s, \left| V^{t+1}(s) - V^\pi(s) \right|$$

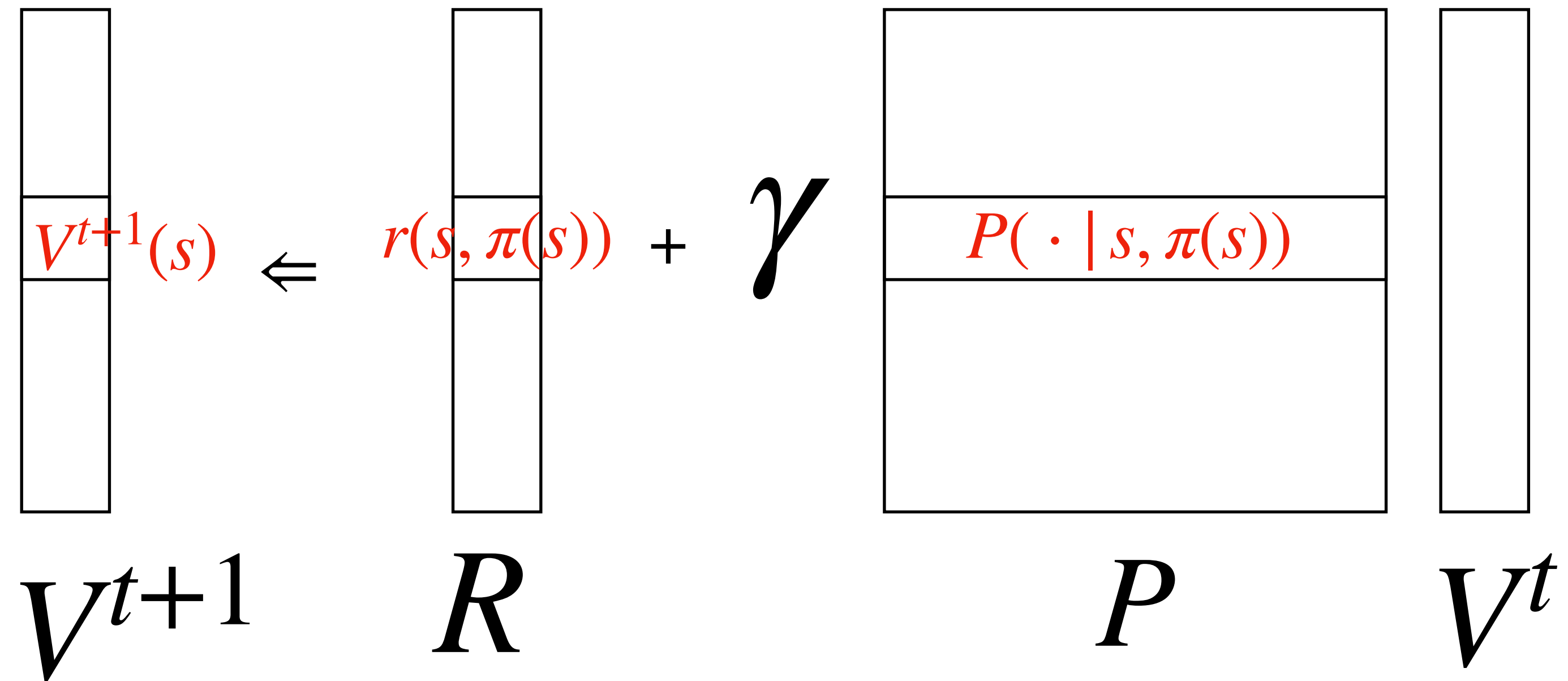
$$= \left| r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \left(r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right) \right|$$

$$= \gamma \left| \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right|$$

$$\leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} \left| V^t(s') - V^\pi(s') \right|$$

$$\leq \gamma \left\| V^t - V^\pi \right\|_\infty \Rightarrow \left\| V^{t+1} - V^\pi \right\|_\infty \leq \gamma \left\| V^t - V^\pi \right\|_\infty$$

Summary so far:



Convergence:

$$\| V^{t+1} - V^\pi \|_\infty \leq \gamma \| V^t - V^\pi \|_\infty \leq \gamma^{t+1} \| V^0 - V^\pi \|_\infty$$

Recap: Optimal Policy $\pi^\star$

For infinite horizon discounted MDP, there always exists a deterministic policy

$$\pi^\star : S \mapsto A, \text{ s.t., } V^{\pi^\star}(s) \geq V^\pi(s), \forall s, \pi$$

[Puterman 94 chapter 6, also see theorem 1.7 in the RL monograph—no need to understand the proof]

Recap: Optimal Policy $\pi^\star$

For infinite horizon discounted MDP, there always exists a deterministic policy

$$\pi^\star : S \mapsto A, \text{ s.t., } V^{\pi^\star}(s) \geq V^\pi(s), \forall s, \pi$$

[Puterman 94 chapter 6, also see theorem 1.7 in the RL monograph—no need to understand the proof]

i.e., $\pi^\star$ dominates any other policy π , everywhere!

Recap: Optimal Policy $\pi^\star$

For infinite horizon discounted MDP, there always exists a deterministic policy

$$\pi^\star : S \mapsto A, \text{ s.t., } V^{\pi^\star}(s) \geq V^\pi(s), \forall s, \pi$$

[Puterman 94 chapter 6, also see theorem 1.7 in the RL monograph—no need to understand the proof]

i.e., $\pi^\star$ dominates any other policy π , everywhere!

We often denote $V^\star, Q^\star$ in short for $V^{\pi^\star}, Q^{\pi^\star}$

Recap: Optimal Policy $\pi^\star$

For infinite horizon discounted MDP, there always exists a deterministic policy

$$\pi^\star : S \mapsto A, \text{ s.t., } V^{\pi^\star}(s) \geq V^\pi(s), \forall s, \pi$$

[Puterman 94 chapter 6, also see theorem 1.7 in the RL monograph—no need to understand the proof]

i.e., $\pi^\star$ dominates any other policy π , everywhere!

We often denote $V^\star, Q^\star$ in short for $V^{\pi^\star}, Q^{\pi^\star}$

$V^\star(s)$: the maximum value we could possibly achieve

Question for Today:

Given an MDP $\mathcal{M} = (S, A, P, r, \gamma)$,
How to find $Q^\star$ and $\pi^\star$ (approximately)

Motivation for Finding the Optimal Policy

Motivation for Finding the Optimal Policy

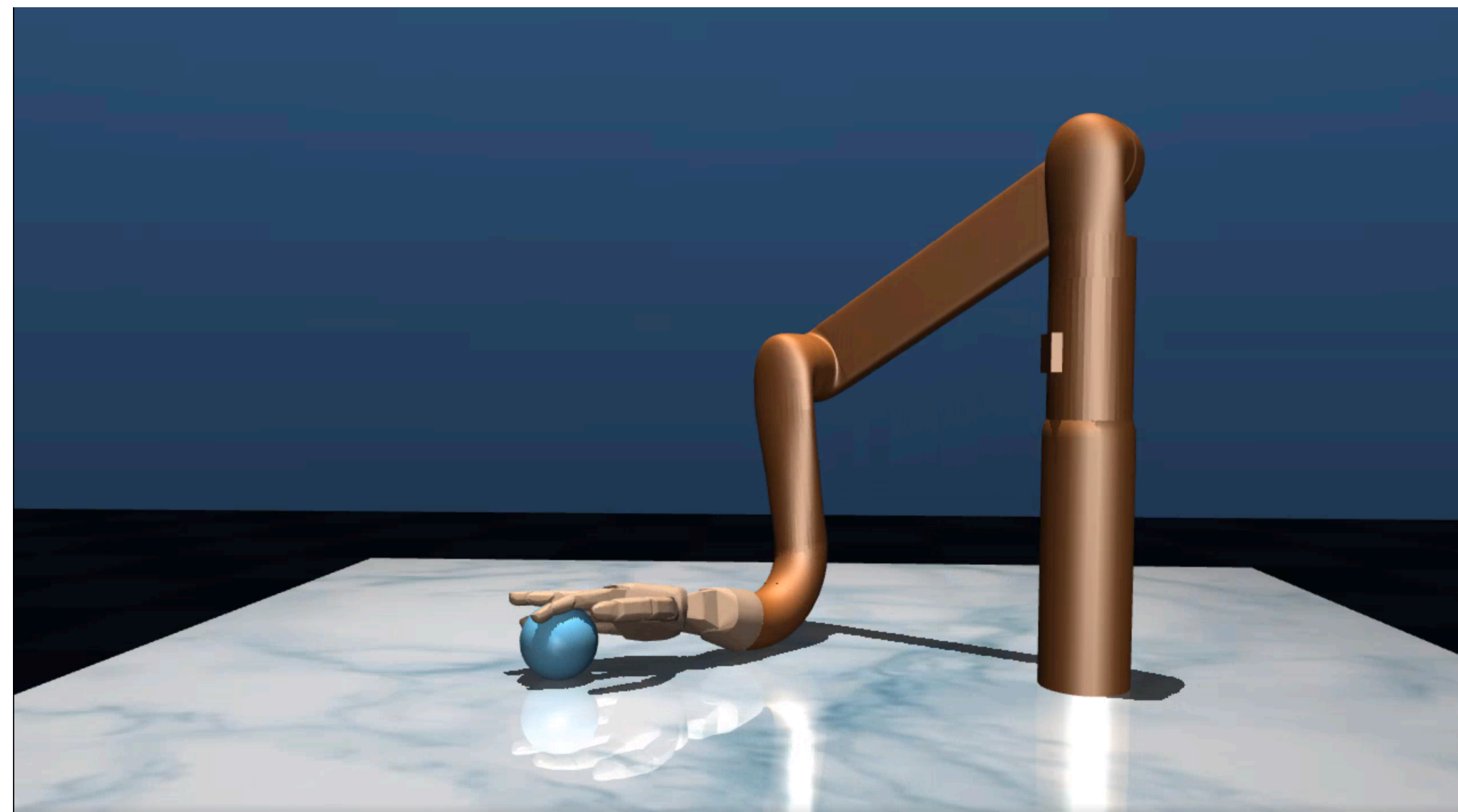


Find the strategy w/ the highest
prob of winning
(i.e., a policy that maps the board
position to the next move)

Motivation for Finding the Optimal Policy



Find the strategy w/ the highest
prob of winning
(i.e., a policy that maps the board
position to the next move)



Find the strategy (i.e., a mapping from
robot & ball configuration to torques)
that picks the ball and moves it to a
goal position ASAP

Outline:

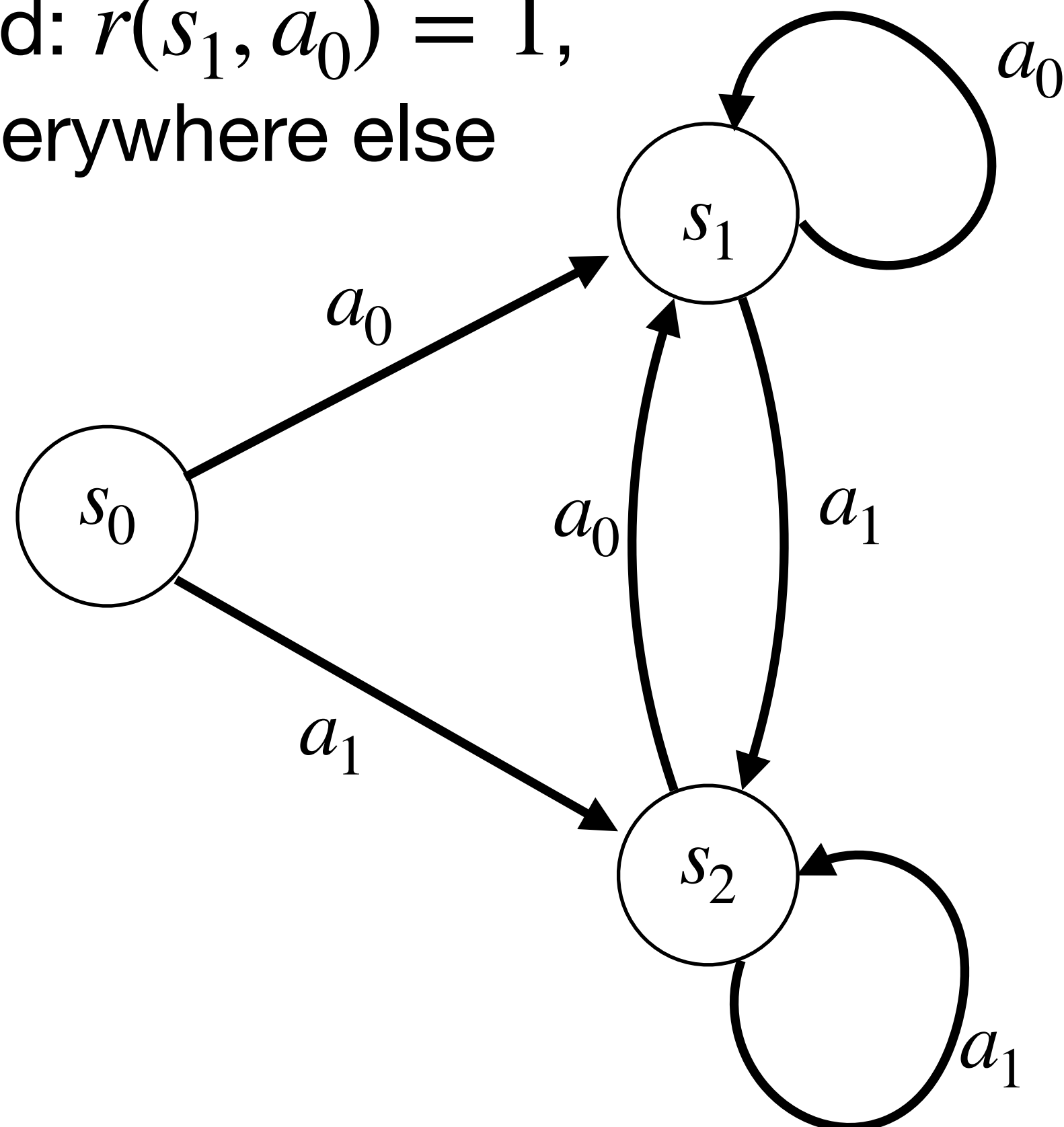
1: Bellman Optimality

2: An Iterative Algorithm: Value Iteration

Example of Optimal Policy $\pi^\star$

Consider the following **deterministic** MDP w/ 3 states & 2 actions

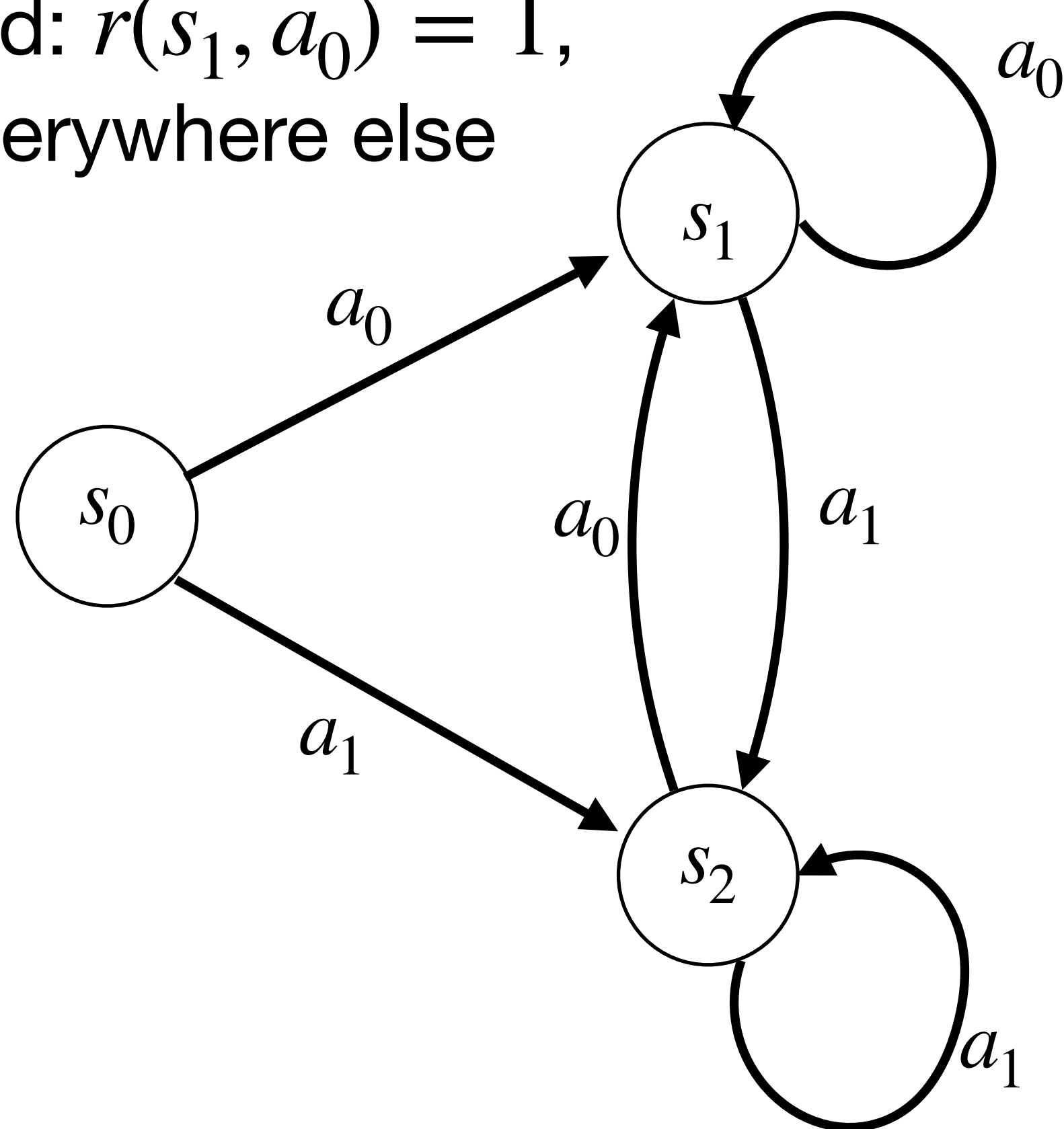
Reward: $r(s_1, a_0) = 1$,
0 everywhere else



Example of Optimal Policy $\pi^\star$

Consider the following **deterministic** MDP w/ 3 states & 2 actions

Reward: $r(s_1, a_0) = 1$,
0 everywhere else

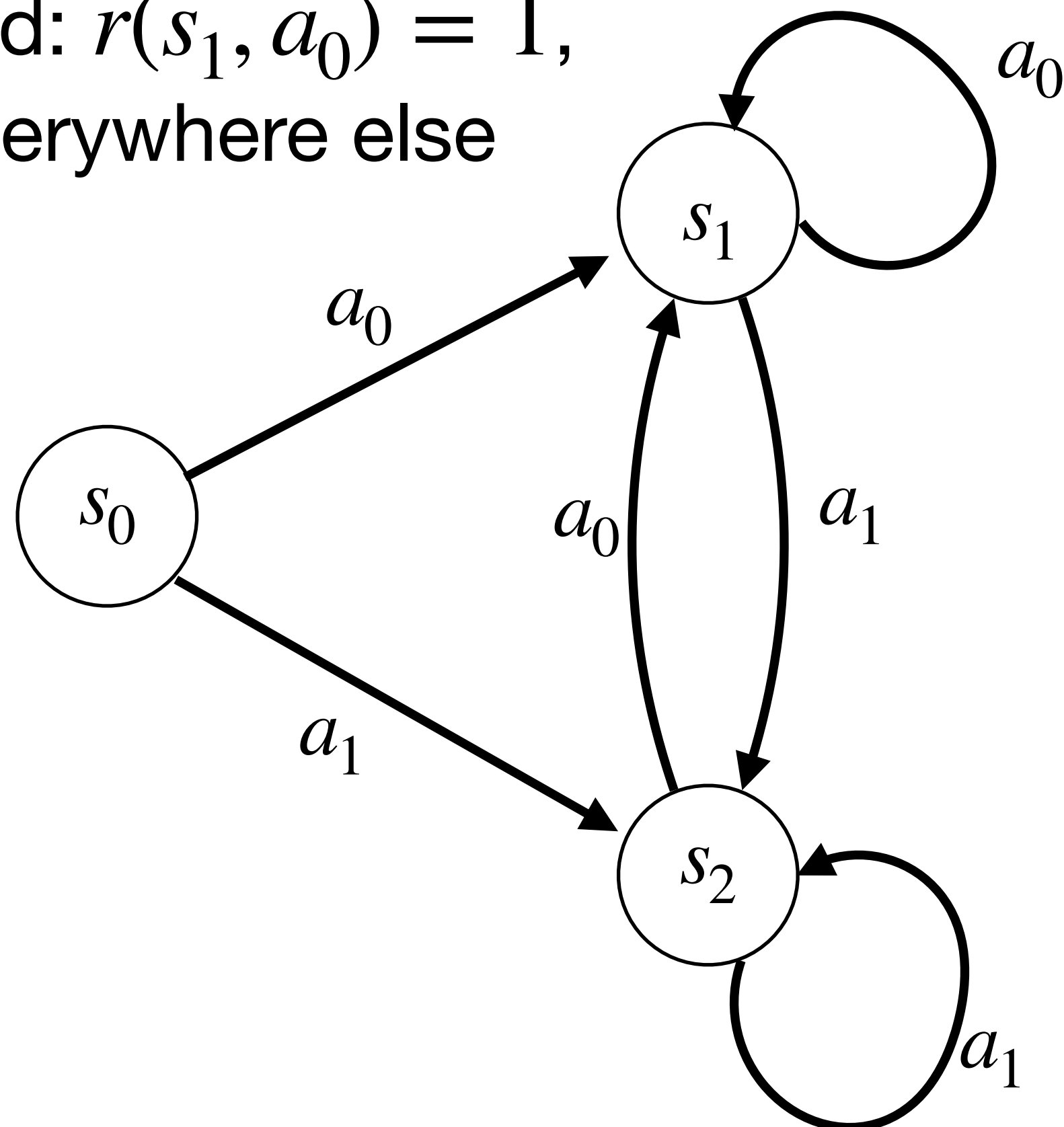


Let's say $\gamma \in (0,1)$
What's the optimal policy?

Example of Optimal Policy $\pi^\star$

Consider the following **deterministic** MDP w/ 3 states & 2 actions

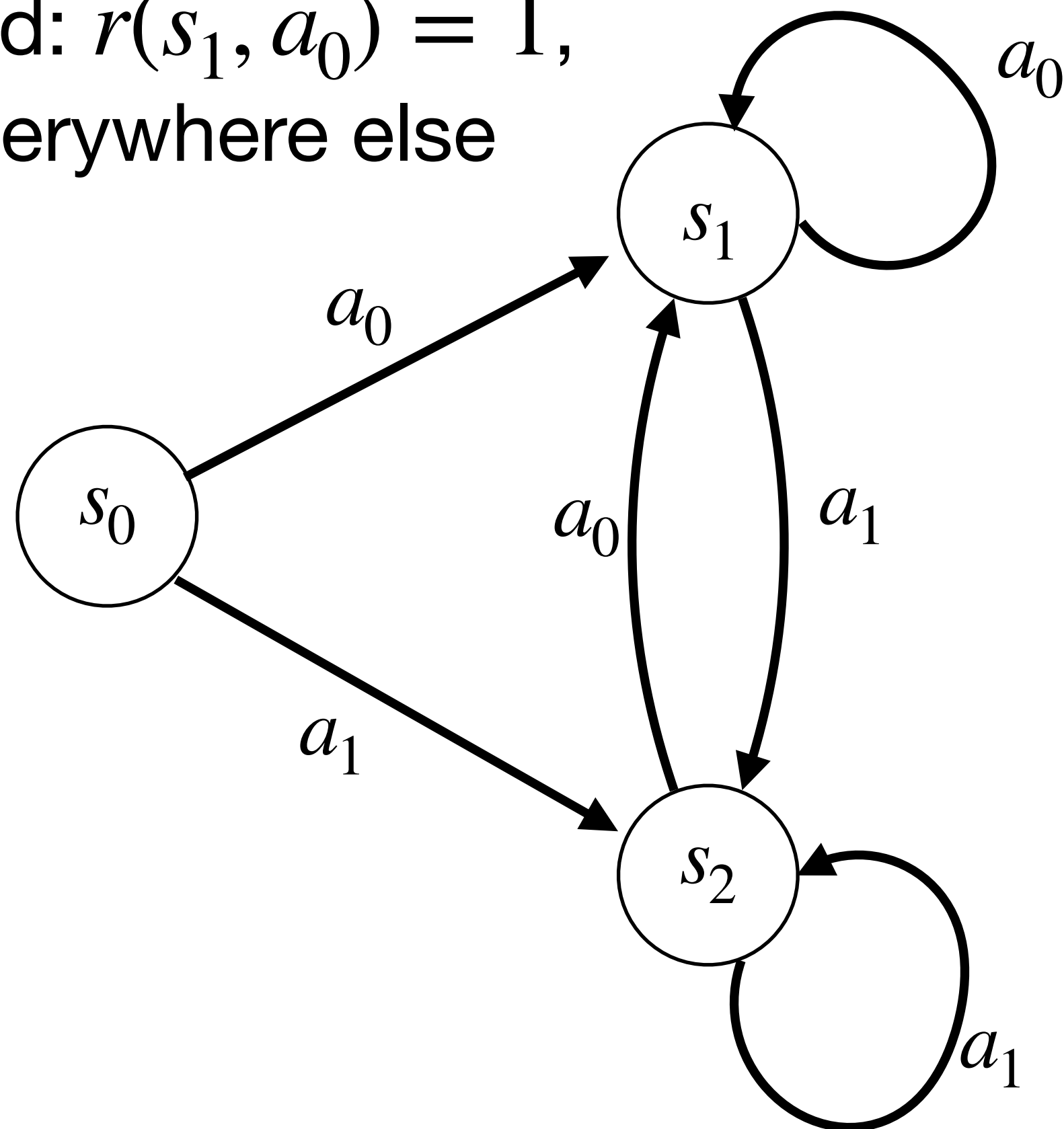
Reward: $r(s_1, a_0) = 1$,
0 everywhere else



Example of Optimal Policy $\pi^\star$

Consider the following **deterministic** MDP w/ 3 states & 2 actions

Reward: $r(s_1, a_0) = 1$,
0 everywhere else



If we were told $V^\star(s_1)$ & $V^\star(s_2)$,
how to compute $V^\star(s_0)$

Bellman Optimality

Bellman Optimality

$$V^*(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^*(s') \right], \forall s$$

Bellman Optimality

Bellman Optimality

$$V^{\star}(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^{\star}(s') \right], \forall s$$

$$V^{\star}(s) = \left[r(s, \pi^{\star}(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi^{\star}(s))} V^{\star}(s') \right] \quad (\text{By BE of } \pi^{\star})$$

Bellman Optimality

Bellman Optimality

$$V^{\star}(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^{\star}(s') \right], \forall s$$

$$\begin{aligned} V^{\star}(s) &= \left[r(s, \pi^{\star}(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi^{\star}(s))} V^{\star}(s') \right] \quad (\text{By BE of } \pi^{\star}) \\ &\leq \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^{\star}(s') \right] \end{aligned}$$

Bellman Optimality

Bellman Optimality

$$V^*(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^*(s') \right], \forall s$$

$$V^*(s) = \left[r(s, \pi^*(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi^*(s))} V^*(s') \right] \quad (\text{By BE of } \pi^*)$$

$$\leq \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^*(s') \right]$$

If we took this $\arg \max_a$ at s , then follow π^* , we **would have higher value**

Bellman Optimality

Bellman Optimality

$$V^*(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^*(s') \right], \forall s$$

$$V^*(s) = \left[r(s, \pi^*(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi^*(s))} V^*(s') \right] \quad (\text{By BE of } \pi^*)$$

$$\leq \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^*(s') \right]$$

If we took this $\arg \max_a$ at s , then follow π^* , we **would have higher value**

Contradicts to the fact that $V^*(s)$ is the maximum value at s one could possibly achieve

Bellman Optimality

Bellman Optimality

$$V^{\star}(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^{\star}(s') \right]$$

Recall that $Q^{\star}(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^{\star}(s')$

Bellman Optimality

Bellman Optimality

$$V^*(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)} V^*(s') \right]$$

Recall that $Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)} V^*(s')$

This implies that $\arg \max_a Q^*(s, a)$ is an optimal policy

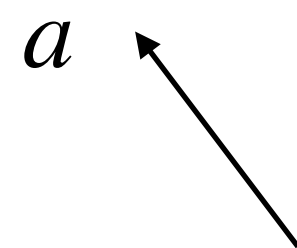
Bellman Optimality

Bellman Optimality

$$V^*(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)} V^*(s') \right]$$

Recall that $Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)} V^*(s')$

This implies that $\arg \max_a Q^*(s, a)$ is an optimal policy



An optimal policy should pick
this action at s

Outline:



1: Bellman Optimality

2: An Iterative Algorithm: Value Iteration

Define the Bellman optimality for $Q^\star$

We now know:

$$V^\star(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^\star(s') \right]$$

What's the Q version?

Define the Bellman optimality for $Q^\star$

We now know:

$$V^\star(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^\star(s') \right]$$

What's the Q version?

$$Q^\star(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} Q^\star(s', a')$$

Define the Bellman optimality for $Q^\star$

We now know:

$$V^\star(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^\star(s') \right]$$

What's the Q version?

$$Q^\star(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} Q^\star(s', a')$$

To estimate $Q^\star$, we will use the fix-point iterative approach again

Define the Bellman operator

Given a function $Q : S \times A \mapsto \mathbb{R}$,

$$\mathcal{T}Q : S \times A \mapsto \mathbb{R},$$

$$(\mathcal{T}Q)(s, a) := r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a' \in A} Q(s', a'), \forall s, a \in S \times A$$

Define the Bellman operator

Given a function $Q : S \times A \mapsto \mathbb{R}$,

$$\mathcal{T}Q : S \times A \mapsto \mathbb{R},$$

$$(\mathcal{T}Q)(s, a) := r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a' \in A} Q(s', a'), \forall s, a \in S \times A$$

We can express $Q \in \mathbb{R}^{|S||A|}$, so $\mathcal{T}Q \in \mathbb{R}^{|S||A|}$

Define the Bellman operator

Given a function $Q : S \times A \mapsto \mathbb{R}$,

$$\mathcal{T}Q : S \times A \mapsto \mathbb{R},$$

$$(\mathcal{T}Q)(s, a) := r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a' \in A} Q(s', a'), \forall s, a \in S \times A$$

We can express $Q \in \mathbb{R}^{|S||A|}$, so $\mathcal{T}Q \in \mathbb{R}^{|S||A|}$

i.e., think about $\mathcal{T}$ as a (non-linear) mapping that maps from $\mathbb{R}^{|S||A|}$ to $\mathbb{R}^{|S||A|}$

High Level idea for Algorithm Design

Fix-point iteration again!

High Level idea for Algorithm Design

Fix-point iteration again!

Recall Bellman Optimality for $Q^\star$:

$$Q^\star(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} Q^\star(s', a')$$

High Level idea for Algorithm Design

Fix-point iteration again!

Recall Bellman Optimality for $Q^\star$:

$$Q^\star(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} Q^\star(s', a')$$

We have $Q^\star = \mathcal{T} Q^\star$,
i.e., $Q^\star$ is a fix-point solution of $Q = \mathcal{T} Q$

Value Iteration Algorithm:

1. Initialization: Q^0

2. Iterate until convergence: $Q^{t+1} \Leftarrow \mathcal{T} Q^t$

Value Iteration Algorithm:

1. Initialization: Q^0

2. Iterate until convergence: $Q^{t+1} \Leftarrow \mathcal{T} Q^t$

We hope $Q^t \rightarrow Q^*$, as $t \rightarrow \infty$

Convergence of Value Iteration:

Lemma [contraction]: Given any Q, Q' , we have:

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_{\infty} \leq \gamma \|Q - Q'\|_{\infty}$$

Proof:

Convergence of Value Iteration:

Lemma [contraction]: Given any Q, Q' , we have:

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_\infty \leq \gamma \|Q - Q'\|_\infty$$

Proof:

$$|(\mathcal{T}Q)(s, a) - (\mathcal{T}Q')(s, a)| = \left| r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q(s', a') - \left(r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q'(s', a') \right) \right|$$

Convergence of Value Iteration:

Lemma [contraction]: Given any Q, Q' , we have:

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_\infty \leq \gamma \|Q - Q'\|_\infty$$

Proof:

$$\begin{aligned} |(\mathcal{T}Q)(s, a) - (\mathcal{T}Q')(s, a)| &= \left| r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q(s', a') - \left(r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q'(s', a') \right) \right| \\ &\leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left| \left(\max_{a'} Q(s', a') - \max_{a'} Q'(s', a') \right) \right| \end{aligned}$$

Convergence of Value Iteration:

Lemma [contraction]: Given any Q, Q' , we have:

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_{\infty} \leq \gamma \|Q - Q'\|_{\infty}$$

Proof:

$$\begin{aligned} |(\mathcal{T}Q)(s, a) - (\mathcal{T}Q')(s, a)| &= \left| r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q(s', a') - \left(r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q'(s', a') \right) \right| \\ &\leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left| \left(\max_{a'} Q(s', a') - \max_{a'} Q'(s', a') \right) \right| \leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} |Q(s', a') - Q'(s', a')| \end{aligned}$$

Convergence of Value Iteration:

Lemma [contraction]: Given any Q, Q' , we have:

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_\infty \leq \gamma \|Q - Q'\|_\infty$$

Proof:

$$\begin{aligned} |(\mathcal{T}Q)(s, a) - (\mathcal{T}Q')(s, a)| &= \left| r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q(s', a') - \left(r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q'(s', a') \right) \right| \\ &\leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left| \left(\max_{a'} Q(s', a') - \max_{a'} Q'(s', a') \right) \right| \leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} |Q(s', a') - Q'(s', a')| \\ &\leq \gamma \max_{s'} \max_{a'} |Q(s', a') - Q'(s', a')| \end{aligned}$$

Convergence of Value Iteration:

Lemma [contraction]: Given any Q, Q' , we have:

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_\infty \leq \gamma \|Q - Q'\|_\infty$$

Proof:

$$\begin{aligned} |(\mathcal{T}Q)(s, a) - (\mathcal{T}Q')(s, a)| &= \left| r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q(s', a') - \left(r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q'(s', a') \right) \right| \\ &\leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left| \left(\max_{a'} Q(s', a') - \max_{a'} Q'(s', a') \right) \right| \leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} |Q(s', a') - Q'(s', a')| \\ &\leq \gamma \max_{s'} \max_{a'} |Q(s', a') - Q'(s', a')| = \gamma \|Q - Q'\|_\infty \end{aligned}$$

Convergence of Value Iteration:

Lemma [Convergence]: Given Q^0 , we have:

$$\|Q^t - Q^*\|_\infty \leq \gamma^t \|Q^0 - Q^*\|_\infty$$

Convergence of Value Iteration:

Lemma [Convergence]: Given Q^0 , we have:

$$\|Q^t - Q^*\|_\infty \leq \gamma^t \|Q^0 - Q^*\|_\infty$$

$$\|Q^{t+1} - Q^*\|_\infty = \|\mathcal{T}Q^t - \mathcal{T}Q^*\|_\infty \leq \gamma \|Q^t - Q^*\|_\infty$$

Convergence of Value Iteration:

Lemma [Convergence]: Given Q^0 , we have:

$$\|Q^t - Q^\star\|_\infty \leq \gamma^t \|Q^0 - Q^\star\|_\infty$$

$$\|Q^{t+1} - Q^\star\|_\infty = \|\mathcal{T}Q^t - \mathcal{T}Q^\star\|_\infty \leq \gamma \|Q^t - Q^\star\|_\infty$$

$$\dots \leq \gamma^{t+1} \|\widehat{Q}^0 - Q^\star\|_\infty$$

Summary so far:

VI (a fix point iteration alg):

$$Q^{t+1} \Leftarrow \mathcal{T} Q^t$$

VI convergence (via contraction)

$$\text{i.e., } \|Q^t - Q^\star\|_\infty \leq \gamma^t \|Q^0 - Q^\star\|_\infty$$

Summary so far:

VI (a fix point iteration alg):

$$Q^{t+1} \Leftarrow \mathcal{T} Q^t$$

VI convergence (via contraction)

$$\text{i.e., } \|Q^t - Q^*\|_\infty \leq \gamma^t \|Q^0 - Q^*\|_\infty$$

What about the policy? Ultimately, we do want π^* ...

